

1. The periodic table is useful for chemists as it arranges the elements in a particular way.
 - (a) What is a group?
 - (b) Why are elements put into a particular group?
 - (c) What are the group numbers of:
 - (i) alkali metals?
 - (ii) inert (noble) gases?
 - (iii) halogens?
 - (d) Where do you find the following elements in the table? Indicate them on the periodic table below, either by labelling or colouring.
 - (i) transition elements
 - (ii) metalloids / semi-metals
 - (e) Where do you find the non-metal elements? Show this on the diagram below.

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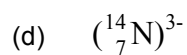
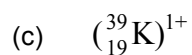
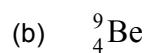
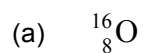
2. Give a definition or description for the following terms.

- (a) atom
- (b) element
- (c) molecule
- (d) compound
- (e) period
- (f) atomic number
- (g) atomic mass
- (h) ion
- (i) valence electron
- (g) valency

3. Complete the following table.

ELEMENT / ION	PROTON NUMBER	NEUTRON NUMBER	ELECTRON NUMBER
$^{24}_{12}\text{Mg}$			
$(^{35}_{17}\text{Cl})^{1-}$			
$^{56}_{26}\text{Fe}$			
$(^{27}_{13}\text{Al})^{3+}$			
$^{16}_8\text{O}$			

4. Draw clear diagrams to show the electron configurations of the following elements.



5. Using $(^{35}_{17}\text{Cl})^{1-}$ as an example, explain the following.

(a) Were electrons gained or lost to form the Cl^{1-} ion?

(b) How many electrons were gained or lost?

(c) In which group is Cl?

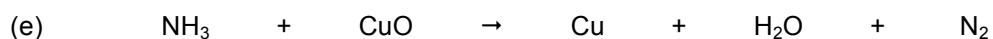
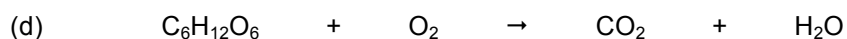
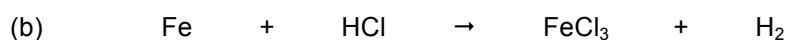
(d) What valency do all members of this group possess?

(e) Explain carefully why the $(^{35}_{17}\text{Cl})^{1-}$ ion is stable, referring to the Octet Rule.
(HINT: Some mention of the noble or inert gases would be appropriate!)

6. Complete the following table.

NAME	FORMULA
	CO ₂
aluminium sulphate	
dinitrogen tetroxide	
calcium hydroxide	
	(NH ₄) ₂ CO ₃
copper (II) chloride	
	NO ₂
potassium nitrate	
	ZnS
sodium phosphate	

7. Balance the following equations.



8. Write balanced equations for any of the following mixtures where a precipitate is produced. Be sure to indicate which substance forms a solid - use a subscript (s). If a reaction does not occur, indicate by writing "no reaction".

(a) Calcium chloride solution + sodium carbonate solution.

(b) Barium hydroxide solution + dilute sulphuric acid.

(c) Sodium nitrate solution + sodium chloride solution.

- (d) Potassium nitrate solution + lead(II) acetate solution.
- (e) Sodium hydroxide solution + calcium chloride solution.
- (f) Ammonium carbonate solution + aluminium nitrate solution.
- (g) Magnesium nitrate solution + sodium carbonate solution.

9. The rate of chemical reactions can be affected by several factors. Use your knowledge of these factors to explain the following.

- (a) Small pieces of wood are used to start a campfire, rather than large logs.
- (b) An iron nail slowly glows red-hot in a Bunsen flame, but steel wool burns quickly.
- (c) You can use 1M HCl to produce H_2 gas by reacting it with Mg metal, but 2 M HCl produces more gas in a shorter time.
- (d) Amylase is a digestive enzyme (catalyst) needed by humans to digest carbohydrates.